

Adapting Point Cloud Analysis via Multimodal Bayesian Distribution Learning

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Abstract

Large multimodal 3D vision–language models show strong generalization across diverse 3D tasks, but their performance still degrades notably under domain shifts. This has motivated recent studies on test-time adaptation (TTA), which enables models to adapt online using test-time data. Among existing TTA methods, cache-based mechanisms are widely adopted for leveraging previously observed samples in online prediction refinement. However, they store only limited historical information, leading to progressive information loss as the test stream evolves. In addition, their prediction logits are fused heuristically, making adaptation unstable. To address these limitations, we propose **BayesMM**, a Multimodal Bayesian Distribution Learning framework for test-time point cloud analysis. BayesMM models textual priors and streaming visual features of each class as Gaussian distributions: textual parameters are derived from semantic prompts, while visual parameters are updated online with arriving samples. The two modalities are fused via Bayesian model averaging, which automatically adjusts their contributions based on posterior evidence, yielding a unified prediction that adapts continually to evolving test-time data without training. Extensive experiments on multiple point cloud benchmarks demonstrate that BayesMM maintains robustness under distributional shifts, yielding over 4% average improvement.

1. Introduction

3D sensors such as LiDAR and RGB-D cameras [6, 9] have become fundamental to robotics and autonomous driving for their reliable geometric perception [17, 28], driving advances in scene reconstruction, object recogni-

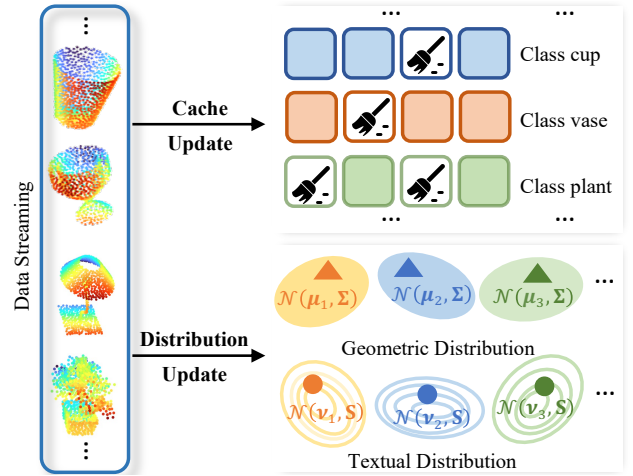


Figure 1. Comparison between cache-based adaptation and our distribution-based modeling. (a) Cache-based methods rely on discrete memory updates, storing a small number of recent samples in a fixed-size cache. (b) Our BayesMM models class-wise distributions across modalities rather than individual samples.

tion, and spatial understanding. Building on this foundation, **large multimodal 3D models** [23, 38, 39] have recently emerged by leveraging contrastive pre-training on large-scale point–image–text triplets. They align geometric and textual representations within a shared embedding space, enabling open-vocabulary point cloud recognition and **strong zero-shot generalization** [16, 48, 51, 53], thus demonstrating the potential of multimodal learning for scalable and generalizable 3D perception [4, 29, 33, 35].

Despite the remarkable progress of large multimodal 3D models, their performance often degrades when facing **domain shifts** between training and testing distributions. Recent studies have thus explored **test-time adaptation (TTA)** [2, 3, 7, 14, 24, 26, 43, 52], enabling models to refine predictions dynamically using unlabeled test data

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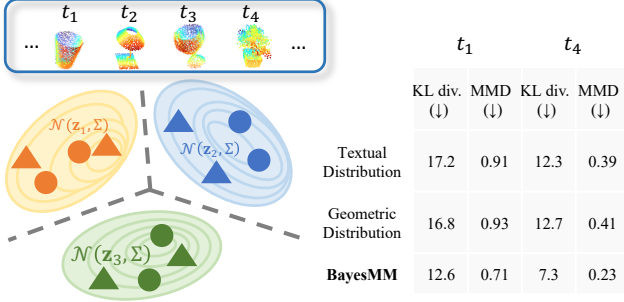


Figure 2. Comparison of distribution consistency across adaptation steps. The Kullback–Leibler (KL) divergence and Maximum Mean Discrepancy (MMD) are measured at different time steps during test-time adaptation.

without retraining. Among these approaches, cache-based methods have shown particular promise by maintaining a compact memory of high-confidence test samples for on-line model adjustment. However, their limited cache capacity causes progressive information loss, failing to capture long-term distributional statistics, as illustrated in the top of Figure 1. As the test stream evolves, continuous sample replacement further amplifies this problem, leading to unstable adaptation and even catastrophic forgetting. In addition, the heuristic fusion between cache-based and zero-shot logits relies on empirically tuned hyperparameters [14, 24, 46], making the adaptation process unstable across domains. These limitations hinder the practicality of cache-based methods in real-world scenarios.

To address the above limitations, we propose BayesMM, a training-free dynamic Bayesian distribution learning framework for adaptive point cloud recognition. As illustrated in the bottom part of Figure 1, BayesMM jointly models textual and geometric modalities under a unified probabilistic formulation. Specifically, it assumes that features of each class follow Gaussian distributions in both modalities. The textual distributions are first derived from semantic embeddings, providing class-wise priors that capture semantic diversity across prompt variants. while the geometric distributions are progressively refined from the test stream to reflect visual variations. The Bayesian formulation allows the model to automatically adjust modality weights, yielding a unified predictive distribution that ensures stable and consistent adaptation.

To quantify the multimodal consistency achieved by BayesMM, we measure the Kullback–Leibler (KL) divergence [12] and Maximum Mean Discrepancy (MMD) [27] between the learned multimodal distributions and their ground-truth references along the adaptation trajectory. As shown in the right part of Figure 2, the full Bayesian fusion attains substantially lower KL and MMD values than its single-modality ablations, indicating more coherent align-

ment between textual and geometric representations. Both metrics steadily decrease as adaptation proceeds, showing that BayesMM continuously refines the joint feature space rather than overfitting to short-term samples. Specifically, the average KL divergence drops from 17.2 to 12.6, and the MMD decreases from 0.91 to 0.71 between the initial and later stages (t_1 – t_4), demonstrating that our Bayesian fusion effectively stabilizes feature dynamics and enhances cross-modal distribution consistency over time.

The main contributions are summarized as follows:

- We propose BayesMM, a training-free dynamic Bayesian distribution learning framework for adaptive point cloud recognition at test time.
- We formulate a unified probabilistic model that jointly models geometric and textual modalities through dynamic parameter updates and Bayesian fusion to achieve cross-modal alignment.
- We conduct extensive experiments on multiple benchmarks, demonstrating that BayesMM achieves robust and consistent test-time adaptation performance across diverse 3D scenarios.

2. Related Work

Test-time adaptation with large multimodal 3D models. TTA [10, 14, 52] aims to address distribution shifts by adapting model representations during inference using test data, without accessing source data. In the 3D domain, recent works such as MATE [18], BFTT3D [30], and CloudFixer [22] explore adaptive strategies for point cloud recognition through masked auto-encoding, prototype memory, and diffusion-based restoration. However, these approaches depend on source-domain data, making them less suitable for TTA. The emergence of large multimodal 3D models [15, 40, 44, 50] has enabled generalizable and open-vocabulary 3D understanding. Representative models such as ULIP-2 [39], OpenShape [16], and Uni3D [48] jointly pre-train on large-scale point–image–text triplets via contrastive alignment [1], unifying geometric and semantic representations for zero-shot generalization. Building on these foundation models, TTA in large multimodal 3D models has recently been explored through cache-based mechanisms [14, 19, 24, 34, 36, 45]. These methods maintain a cache of feature representations collected during inference and retrieve relevant entries to guide prediction updates, enabling efficient on-the-fly adaptation. In contrast, our BayesMM formulates test-time adaptation as dynamic multimodal distribution learning. It continuously models geometric distributions and integrates them with textual priors via Bayesian inference, where modality weights are automatically adjusted under the Bayesian principle, enabling robust cloud recognition under distribution shifts.

t_1 and t_4 correspond to the 500th and 2000th test samples on ModelNet-C [21], respectively.

Distribution learning. Distribution learning provides a principled framework for adapting recognition models by exploiting the statistical structure of feature space rather than relying on fixed representations. Classical approaches such as Gaussian Discriminant Analysis [11, 31, 51] assume that features of each class follow Gaussian distributions and construct probabilistic classifiers in closed form. Recent advances extend this idea to test-time scenarios. DOTA [10] formulates test-time adaptation of vision–language models as online estimation of Gaussian parameters from test data streams to capture non-stationary shifts. Online Gaussian Test-Time Adaptation [8, 42] further updates class-wise means and covariances during inference for continual adaptation without external memory. More recently, ADAPT [47] aligns Gaussian-distributed test features with class prototypes by adjusting per-class statistics, while BCA [49] incrementally refines Gaussian parameters using incoming samples for efficient and stable source-free adaptation. Different from these unimodal Gaussian-based approaches, BayesMM performs multimodal distribution learning by jointly estimating geometric and textual distributions and integrating them through Bayesian inference for robust point cloud recognition.

3. Methodology

In this section, we present BayesMM, a test-time adaptation framework that models geometric and textual modalities as evolving distributions for robust 3D recognition. An overview is shown in Figure 3.

3.1. Setup

We consider a streaming test-time scenario for large multimodal 3D data [24, 38, 39, 44, 50], where a sequence of point clouds $\{X_t\}_{t=1}^{\infty}$ arrives online, accompanied by a fixed set of text prototypes $\{T_c\}_{c=1}^C$ (e.g., “a 3D object of [classname]_c”). A *frozen* point encoder Φ and a *frozen* text encoder Ψ project the inputs into a shared feature space:

$$\mathbf{x}_t = \Phi(X_t) \in \mathbb{R}^d, \quad \mathbf{z}_c = \Psi(T_c) \in \mathbb{R}^d.$$

On top of these fixed embeddings, a lightweight head $f_{\theta_t} : \mathbb{R}^d \rightarrow \mathbb{R}^C$ produces prediction scores, where the parameters θ_t are updated online as new test samples arrive.

As an example, we illustrate the cache-based test-time adaptation strategy [24]. At the initial time ($t = 0$), the classifier reduces to the zero-shot one, whose parameters are given by the text prototypes $\theta_0 = \{\mathbf{z}_c\}_{c=1}^C$. For a sample $\mathbf{x}_0$, the class score is computed as:

$$f_{\theta_0}(\mathbf{x}_0)_c = \mathbf{z}_c^\top \mathbf{x}_0. \quad (1)$$

At time step t , the model maintains a class-wise cache $\mathbf{h}_{t,c}$ that stores up to K historical embeddings of test samples predicted with high confidence as class c . The parameters

at time t are thus $\theta_t = \{\mathbf{z}_c, \mathbf{h}_{t,c}\}_{c=1}^C$. Given a new test sample $\mathbf{x}_t$, the scoring function combines the text similarity and the cache similarity:

$$f_{\theta_t}(\mathbf{x}_t)_c = \mathbf{z}_c^\top \mathbf{x}_t + \lambda \exp(-\gamma[1 - \cos(\mathbf{x}_t, \mathbf{h}_{t,c})]), \quad (2)$$

where $\lambda > 0$ balances the contributions of the zero-shot prototype and the cached features, and γ controls the sensitivity of cosine distance.

3.2. Multimodal distribution learning

Cache-based adaptation [14, 24, 46] suffers from two issues: limited cache capacity causes information decay, and heuristic logit fusion based on empirical hyperparameters (e.g., λ, γ in Eq. (2)) lacks a theoretical principle. In contrast, our BayesMM models textual and geometric modalities as distributional representations and fuses their classification results under a Bayesian model averaging formulation, effectively utilizing information from previous samples during continuous updates.

Textual distribution learning. To establish reliable semantic priors, we first construct textual distributions capturing the semantic diversity across classes. For each class c , the base prompt “a 3D object of {class}” is expanded by an LLM into M paraphrases, producing embeddings $\{\mathbf{z}^{c,1}, \dots, \mathbf{z}^{c,M}\}$ that reflect varied conceptual descriptions of the same category. The empirical mean and covariance of class c are then computed as:

$$\bar{\mathbf{z}}^c = \frac{1}{M} \sum_{i=1}^M \mathbf{z}^{c,i}, \quad \mathbf{S}^c = \sum_{i=1}^M (\mathbf{z}^{c,i} - \bar{\mathbf{z}}^c)(\mathbf{z}^{c,i} - \bar{\mathbf{z}}^c)^\top. \quad (3)$$

We model each textual prototype ν^c as a Gaussian variable centered at the empirical mean $\bar{\mathbf{z}}^c$, reflecting the uncertainty of language representations across prompt variants:

$$p(\nu^c) = \mathcal{N}(\nu^c \mid \bar{\mathbf{z}}^c, \beta^2 \mathbf{I}), \quad (4)$$

where β controls prior variance. Given a test feature $\mathbf{x}_t$ at time t , its likelihood under class c is:

$$p(\mathbf{x}_t \mid \nu^c, \mathbf{S}^c) = \mathcal{N}(\mathbf{x}_t \mid \nu^c, \mathbf{S}^c), \quad (5)$$

where $\mathbf{S}^c$ represents the intra-class variability of textual embeddings. In practice, a shared covariance $\mathbf{S}$ is used for all classes, which is equivalent to imposing a Dirac prior $p(\mathbf{S}^c) = \delta(\mathbf{S}^c - \mathbf{S})$ that treats the covariance as a fixed parameter. By combining the prior and likelihood, the posterior over textual parameters is obtained as:

$$p(\nu^c, \mathbf{S}^c \mid \mathbf{x}_t) \propto p(\mathbf{x}_t \mid \nu^c, \mathbf{S}^c) p(\nu^c), \quad (6)$$

which integrates the semantic evidence from both the textual prior distribution and the incoming visual observation.

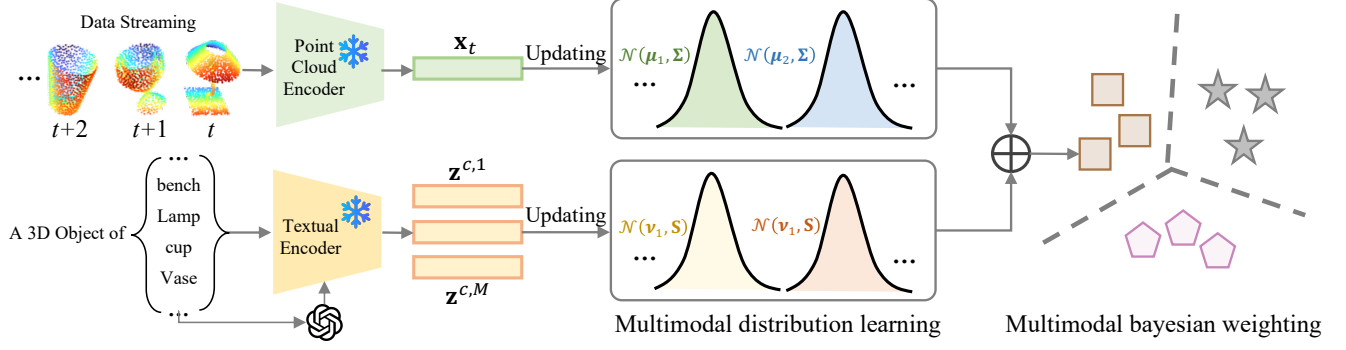


Figure 3. Overview of the proposed BayesMM framework. A frozen point cloud encoder extracts geometric features from streaming inputs, and a frozen language model provides textual embeddings as semantic priors. Both modalities are represented by Gaussian distributions, where geometric ones are updated online with incoming samples. Bayesian weighting fuses the two modalities into a unified posterior for adaptive and training-free point cloud recognition.

The deterministic textual prototype used for inference is then derived via Maximum A Posteriori (MAP) estimation:

$$\nu_{\text{MAP}}^c = (\beta^{-2}\mathbf{I} + M(\mathbf{S}^c)^{-1})^{-1}(\mathbf{S}^c)^{-1}\bar{\mathbf{z}}^c. \quad (7)$$

The detailed derivation of Eq. (7) is provided in Section B of the Supplementary Material.

Geometric distribution learning. With the textual distributions providing semantic priors, we now model an online geometric distribution for each class during test-time, parameterized as a Gaussian set: $\Theta_t^c = \{\mu_t^c, \Sigma_t^c\}$, which is updated sequentially as new samples arrive. At the initial moment ($t=0$), the prior of each class distribution is anchored by its textual prototype $\bar{\mathbf{z}}^c$:

$$p(\mu_0^c) = \mathcal{N}(\mu_0^c | \bar{\mathbf{z}}^c, \alpha^2\mathbf{I}), \quad \Sigma_0^c = \mathbf{S}^c, \quad (8)$$

where α controls the prior variance. The covariance $\mathbf{S}^c$ in Eq. (6), provides an initial estimate of intra-class variability and serves as a semantic prior for geometric adaptation.

At time t , given the new observation $\mathbf{x}_t$, the prior is defined as the previous posterior:

$$p(\Theta_t^c) = p(\Theta_{t-1}^c | \mathbf{x}_{t-1}). \quad (9)$$

The likelihood of observing $\mathbf{x}_t$ under class c is defined as:

$$p(\mathbf{x}_t | \Theta_t^c) = \mathcal{N}(\mathbf{x}_t | \mu_t^c, \Sigma_t^c). \quad (10)$$

Combining the prior and likelihood, the posterior is recursively updated by Bayes' rule:

$$\begin{aligned} p(\Theta_t^c | \mathbf{x}_t) &\propto p(\mathbf{x}_t | \Theta_t^c)p(\Theta_t^c) \\ &\propto p(\mathbf{x}_t | \Theta_t^c)p(\Theta_{t-1}^c | \mathbf{x}_{t-1}). \end{aligned} \quad (11)$$

Under Gaussian assumptions, this recursive update admits a closed-form solution:

$$\begin{aligned} \mu_t^c &= \Sigma_t^c \left((\Sigma^c)^{-1}\mathbf{x}_t + (\Sigma_{t-1}^c)^{-1}\mu_{t-1}^c \right), \\ \Sigma_t^c &= \left((\Sigma_{t-1}^c)^{-1} + (\Sigma^c)^{-1} \right)^{-1}. \end{aligned} \quad (12)$$

The detailed derivation of Eq. (12) is provided in Section B of the Supplemental Material.

3.3. Multimodal bayesian weighting

With modality-specific posteriors available, we fuse geometry and text under Bayesian model averaging [20]. Let $\Omega = \{(\nu^c, \mathbf{S}^c)\}_{c=1}^C$ and $\Theta_t = \{(\mu_t^c, \Sigma_t^c)\}_{c=1}^C$ denote the class-wise parameter sets for the textual and geometric modalities, respectively. The overall posterior for class c at time t is:

$$\begin{aligned} p(c | \mathbf{x}_t) &= \underbrace{p(c | \mathbf{x}_t, \Omega^c) p(\Omega^c | \mathbf{x}_t)}_{\text{Textual posterior predictive (predictive} \times \text{evidence)}} \\ &+ \underbrace{p(c | \mathbf{x}_t, \Theta_t^c) p(\Theta_t^c | \mathbf{x}_t)}_{\text{Geometric posterior predictive (predictive} \times \text{evidence)}} \end{aligned} \quad (13)$$

Each term represents a modality-specific posterior predictive, where $p(\Omega^c | \mathbf{x}_t)$ and $p(\Theta_t^c | \mathbf{x}_t)$ serve as Bayesian weights that automatically balance the two modalities. Under the textual and geometric distributions derived above, we compute their class-conditional posteriors under Gaussian discriminant analysis (GDA). Each modality yields a normalized Gaussian posterior over class c as:

$$\begin{aligned} p(c | \mathbf{x}_t, \Omega^c) &= \frac{\mathcal{N}(\mathbf{x}_t | \nu_{\text{MAP}}^c, \mathbf{S}^c)}{\sum_{c'} \mathcal{N}(\mathbf{x}_t | \nu_{\text{MAP}}^{c'}, \mathbf{S}^{c'})}, \\ p(c | \mathbf{x}_t, \Theta_t^c) &= \frac{\mathcal{N}(\mathbf{x}_t | \mu_t^c, \Sigma_t^c)}{\sum_{c'} \mathcal{N}(\mathbf{x}_t | \mu_t^{c'}, \Sigma_t^{c'})}. \end{aligned} \quad (14)$$

Substituting Eq. (14) into the Eq. (13) yields the final multimodal posterior $p(c | \mathbf{x}_t)$.

4. Experiments

4.1. Experimental settings

Datasets. To evaluate the *robustness* of point cloud recognition, we adopt four public datasets covering diverse corruption types. Specifically, we use ModelNet-C [21] and three

Table 1. Recognition accuracy comparison on ModelNet-C with 7 corruption types. Each clean point cloud contains 1024 points, and the corruption severity level is set to 2. The last column reports the average accuracy over all types. The best results are highlighted in **bold**, and the second best are underlined.

Method	Clean Data	Corruption Type							Avg.
	ModelNet	Add Global	Add Local	Drop Global	Drop Local	Rotate	Scale	Jitter	
ULIP [38]	56.16	33.55	43.92	54.70	50.89	55.27	50.20	44.08	48.60
+ Point-Cache (Global)	62.12	45.79	<u>47.98</u>	56.85	53.89	60.25	54.34	48.91	53.77
+ Point-Cache (Hierarchical)	<u>64.22</u>	<u>46.15</u>	47.85	<u>59.16</u>	<u>56.00</u>	<u>61.47</u>	<u>55.35</u>	<u>49.92</u>	<u>55.02</u>
+ BayesMM	66.04	54.82	53.93	63.09	60.13	63.82	60.49	53.04	59.42
ULIP-2 [39]	71.23	65.15	54.62	68.76	57.98	70.30	67.10	21.76	59.61
+ Point-Cache (Global)	73.95	67.02	59.32	71.35	61.59	72.37	68.40	28.20	62.78
+ Point-Cache (Hierarchical)	<u>74.53</u>	<u>68.11</u>	<u>61.26</u>	73.22	<u>63.65</u>	<u>73.34</u>	<u>70.42</u>	29.50	<u>64.25</u>
+ BayesMM	76.30	69.04	64.38	<u>72.57</u>	64.18	74.55	71.64	<u>29.01</u>	65.21
O-Shape [16]	<u>84.56</u>	71.64	67.79	81.56	73.58	82.01	78.48	59.36	74.87
+ Point-Cache (Global)	84.52	74.72	72.77	82.41	75.12	83.18	78.93	67.91	77.45
+ Point-Cache (Hierarchical)	84.04	<u>74.84</u>	<u>73.70</u>	<u>82.21</u>	<u>76.26</u>	<u>82.66</u>	78.12	<u>68.35</u>	<u>77.52</u>
+ BayesMM	85.49	75.36	74.39	83.14	77.07	84.08	79.86	69.45	78.61
Uni3D [48]	81.81	72.45	56.36	68.15	67.18	79.94	75.36	56.24	69.69
+ Point-Cache (Global)	83.14	76.13	66.49	71.43	69.81	81.52	75.85	61.43	73.20
+ Point-Cache (Hierarchical)	<u>83.87</u>	<u>77.51</u>	<u>71.15</u>	<u>72.16</u>	<u>70.75</u>	<u>81.77</u>	<u>77.31</u>	<u>62.52</u>	<u>74.63</u>
+ BayesMM	85.17	77.59	73.30	74.96	71.88	83.75	79.98	65.84	76.56

Table 2. Recognition accuracy comparison on multiple benchmarks. S-PB_RS_T50 denotes the hardest split of ScanObjectNN. O-LVIS and Omni3D refer to Objaverse-LVIS and OmniObject3D, respectively. The number under each dataset indicates the number of points per object (pts). In Omni3D, each object may contain a variable number of points.

Method	ModelNet40	S-PB_RS_T50	O-LVIS	Omni3D			Avg.
	(10000 pts)	(2048 pts)	(10000 pts)	(1024 pts)	4096 pts	16384 pts)	
ULIP [38]	58.75	46.44	6.24	8.39	7.75	7.28	22.48
+ Point-Cache (Global)	61.22	50.21	<u>7.02</u>	10.00	9.36	8.43	24.37
+ Point-Cache (Hierarchical)	<u>62.93</u>	<u>51.80</u>	<u>7.02</u>	<u>10.47</u>	<u>9.75</u>	<u>8.90</u>	<u>25.15</u>
+ BayesMM	67.13	53.67	7.79	11.27	10.68	9.47	26.67
ULIP-2 [39]	72.97	47.13	30.26	26.36	29.20	26.58	38.75
+ Point-Cache (Global)	74.51	51.70	<u>32.65</u>	28.51	31.10	28.53	41.17
+ Point-Cache (Hierarchical)	<u>75.53</u>	<u>54.98</u>	32.36	29.37	<u>31.24</u>	<u>29.44</u>	<u>42.15</u>
+ BayesMM	76.78	56.47	32.76	<u>28.98</u>	31.76	31.12	42.98
O-Shape [16]	84.52	54.60	<u>46.78</u>	33.21	33.52	33.37	47.67
+ Point-Cache (Global)	85.70	<u>57.13</u>	47.03	<u>36.92</u>	37.61	<u>37.43</u>	<u>50.30</u>
+ Point-Cache (Hierarchical)	85.90	56.61	45.63	36.87	<u>38.02</u>	37.39	50.07
+ BayesMM	<u>85.74</u>	66.12	43.93	37.77	38.03	38.38	51.66
Uni3D [48]	88.41	65.19	55.42	31.52	41.98	41.86	54.09
+ Point-Cache (Global)	88.86	<u>68.51</u>	53.36	34.97	45.13	45.19	56.00
+ Point-Cache (Hierarchical)	<u>89.18</u>	68.24	<u>55.19</u>	<u>35.82</u>	<u>45.60</u>	<u>45.89</u>	<u>56.65</u>
+ BayesMM	90.48	73.04	53.63	36.54	45.97	46.68	57.72

corrupted variants of ScanObjectNN-C [25]. ModelNet-C defines seven atomic corruptions, including *global outliers*, *local outliers*, *global structure dropping*, *local part dropping*, *rotation*, *scaling*, and *jittering*, from which other corruption types can be derived. Following [21], we apply these atomic corruptions to the three variants of ScanOb-

jectNN to construct their corrupted versions. To assess *generalization* on unseen data, we further test our method on four challenging benchmarks: OmniObject3D [32] (216 classes), Objaverse-LVIS [5] (1,156 classes), the hardest variant of ScanObjectNN, and ModelNet40 [37]. Recognition accuracy (%) is reported as the main metric.

Table 3. Recognition accuracy comparison on Sim-to-Real. Two evaluation settings are considered: MN_11 $\rightarrow$ SONN_11 and SN_9 $\rightarrow$ SONN_9. The dataset on the left side of $\rightarrow$ stands for simulated data, while the dataset on the right side indicates real-world data. 11 classes are shared between MN_11 and SONN_11, while 9 classes are common between SN_9 and SONN_9. In the experiments, each point cloud is represented by 2,048 points. MN: ModelNet, SN: ShapeNet.

Method	MN_11 $\rightarrow$ SONN_11			SN_9 $\rightarrow$ SONN_9			Avg.
	OBJ	OBJ_BG	PB_T50_RS	OBJ	OBJ_BG	PB_T50_RS	
ULIP [38]	57.05	50.32	32.60	61.00	61.00	44.38	51.06
+ Point-Cache (Global)	62.32	52.63	34.97	<u>65.50</u>	62.50	47.36	54.21
+ Point-Cache (Hierarchical)	<u>64.42</u>	<u>56.63</u>	<u>35.77</u>	67.25	<u>64.50</u>	<u>47.61</u>	<u>56.03</u>
+ BayesMM	65.41	60.42	45.34	67.25	68.50	55.36	60.38
ULIP-2 [39]	50.94	52.42	39.12	51.50	59.25	46.35	49.93
+ Point-Cache (Global)	55.10	58.52	47.38	56.75	<u>65.00</u>	50.68	55.57
+ Point-Cache (Hierarchical)	<u>57.26</u>	<u>58.95</u>	<u>47.71</u>	<u>58.00</u>	70.25	<u>52.70</u>	<u>57.48</u>
+ BayesMM	57.89	58.97	50.25	61.25	64.25	56.22	58.14
O-Shape [16]	59.78	62.53	45.51	64.00	70.25	53.55	59.27
+ Point-Cache (Global)	65.07	68.67	46.23	71.00	<u>71.50</u>	<u>55.67</u>	<u>63.02</u>
+ Point-Cache (Hierarchical)	<u>66.11</u>	<u>69.68</u>	<u>47.50</u>	<u>71.50</u>	71.00	56.57	63.78
+ BayesMM	69.05	71.76	56.94	71.75	74.50	64.00	68.00
Uni3D [48]	72.63	74.53	55.76	67.50	68.50	57.98	66.18
+ Point-Cache (Global)	76.21	77.26	<u>59.10</u>	<u>74.50</u>	<u>76.50</u>	<u>62.47</u>	71.01
+ Point-Cache (Hierarchical)	74.11	76.00	57.92	77.50	78.00	58.89	69.07
+ BayesMM	<u>74.31</u>	<u>77.05</u>	62.48	72.50	76.00	63.32	<u>70.94</u>

Models. We evaluate our approach on four representative multimodal 3D foundation models: ULIP [38], ULIP-2 [39], OpenShape [16], and Uni3D [48]. All models are initialized with publicly released pre-trained weights and remain *frozen* during evaluation. For fair comparison, each model operates on point clouds uniformly sampled to 1,024 points and normalized within the unit sphere. All experiments are implemented under a unified evaluation framework, ensuring consistent preprocessing and input configurations. Further implementation details can be found in Section A of the Supplementary Material.

4.2. Main results

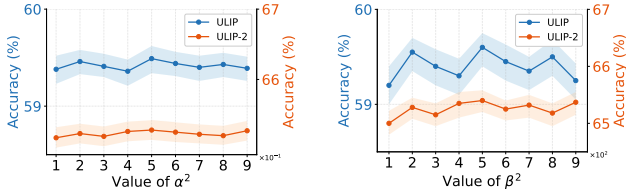
Robustness. We evaluate the test-time robustness of our method on ModelNet-C [21], which includes seven corruption types in addition to the clean set. As shown in Table 1, our training-free BayesMM significantly enhances the robustness of all four large multimodal 3D backbones. When averaged over the clean and corrupted settings, BayesMM improves the performance of ULIP [38] from 48.60% to 59.42% (+10.82), ULIP-2 [39] from 59.61% to 65.21% (+5.60), OpenShape [16] from 74.87% to 78.61% (+3.74), and Uni3D [48] from 69.69% to 76.56% (+6.87). Compared with Point-Cache baselines, BayesMM consistently achieves the highest average accuracy across all corruption types, demonstrating stronger robustness against structural perturbations, outliers, rotations, scale variations, and geometric distortions.

Notably, our BayesMM not only mitigates degradation on corrupted data but also improves recognition on clean inputs. For instance, ULIP and ULIP-2 gain +9.88% and +5.07% absolute improvements on the clean ModelNet dataset, respectively. While OpenShape retains the highest clean accuracy among all models, our cache mechanism yields larger robustness margins under corruption, outperforming its original model by +5.12% on average. Similar consistent gains are observed across all corruption types and across different 3D backbones, demonstrating that our hierarchical caching scheme generalizes well across architectures without any retraining or fine-tuning. Further results on ScanObjectNN-C variants are provided in Table G, H, and I in the Supplementary.

Generalization. To further evaluate the generalization ability of our method under domain and distribution shifts, we test it across four benchmarks: ModelNet40 [37], the hardest split of ScanObjectNN (S-PB_RS_T50) [25], Objaverse-LVIS (O-LVIS) [5], and OmniObject3D [32], as summarized in Table 2. These datasets span diverse object categories, point densities, and real-world variations, providing a comprehensive evaluation of model robustness and transferability. BayesMM consistently improves recognition accuracy across all datasets and architectures without any retraining or fine-tuning. For ULIP [38], the average accuracy rises from 22.48% to 26.67% (+4.19), with similar gains for ULIP-2 [39] (+4.23), OpenShape [16] (+3.99) and Uni3D [48] (+3.63). On the Omni3D dataset, BayesMM

Table 4. Ablation study on the components of BayesMM. S-OBJ denotes the OBJ_ONLY split of ScanObjectNN.

	Geometric Distribution	Textual Distribution	Bayes Weighting	ULIP2 [39]	
				S-OBJ	Omni3D
(1)	✗	✗	✗	42.00	26.58
(2)	✓	✗	✗	46.47	26.63
(3)	✗	✓	✗	<u>52.50</u>	<u>30.79</u>
(4)	✓	✓	✓	53.02	31.12

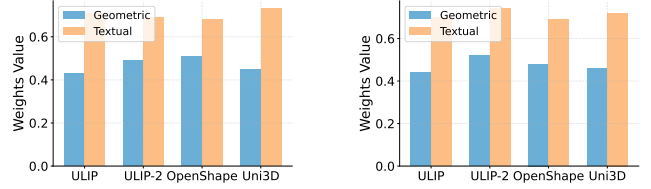


(a) Effect of α^2 on geometric distribution learning (b) Effect of β^2 on textual distribution learning

Figure 4. Hyperparameter sensitivity of BayesMM with respect to the geometric prior α^2 and textual prior β^2 , evaluated on ULIP and ULIP-2 backbones.

exhibits strong generalization across varying point densities (1024, 4096, 16384), achieving substantial performance gains under all settings. Even when the baselines already achieve strong performance (e.g., Uni3D on ModelNet40 and the hardest split of ScanObjectNN), BayesMM continues to deliver steady improvements.

Generalization from simulated to real data. Following Sim-to-Real [13], we further evaluate our method under the cross-domain setting from simulated to real-world data, covering two scenarios: MN.11 $\rightarrow$ SONN.11 and SN.9 $\rightarrow$ SONN.9. As shown in Table 3, BayesMM consistently improves recognition performance across all foundation backbones. Compared with Point-Cache [24], BayesMM ranks at or near the top in almost all metrics, demonstrating stronger generalization of pre-trained 3D models without any additional training. On ULIP [38], our method achieves an average accuracy of 60.38%, outperforming the hierarchical Point-Cache by +4.35%. Consistent gains are also observed on ULIP-2 [39] (+0.66%), OpenShape [16] (+4.22%), and Uni3D [48] (+1.87%), indicating that the proposed distribution learning strategy generalizes robustly across architectures of varying capacity. Notably, BayesMM achieves clear gains on the PB_T50_RS subsets, reflecting its enhanced robustness to real-world noise, background clutter, and geometric perturbations.



(a) Weight comparison on clean data

(b) Weight comparison on corruption data

Figure 5. Analysis of geometric and textual weights in BayesMM across different backbones on the ModelNet-C dataset.

4.3. Ablation study

Ablation on model components. To investigate the contribution of each component within BayesMM, we perform an ablation study on the ULIP2 [38] backbone, evaluated on the ScanObjectNN (OBJ_ONLY split, denoted as S-OBJ) and Omni3D benchmarks, as summarized in Table 4. The baseline using only fixed text prototypes performs poorly due to the lack of adaptive refinement. Introducing geometric distribution learning significantly improves accuracy, as the recursive update in Eq. (12) enables online prototype refinement. Adding textual distribution learning further boosts performance by modeling intra-class semantic variability with LLM-generated paraphrases. Combining both modalities under the Bayesian model averaging principle (Eq. (13)) achieves the best results, showing that geometric and textual distributions are complementary—geometric modeling enhances adaptability, textual modeling enriches semantics, and Bayesian weighting integrates them coherently within a unified probabilistic framework.

Sensitivity analysis of α^2 and β^2 . To examine the sensitivity of BayesMM to hyperparameter choices, α^2 and β^2 are varied to control the weighting strength in Bayesian fusion and the variance scale in multimodal distribution learning, respectively. As shown in Figure 4(a), varying α^2 has minimal impact on accuracy. Both ULIP [38] and ULIP-2 [39] remain stable, indicating that the geometric prior is insensitive to moderate changes in weighting strength and that Bayesian fusion effectively balances geometric and textual cues. Figure 4(b) shows a similar trend for β^2 , where accuracy remains nearly constant across different variance scales. ULIP [38] stays around 59.4%, while ULIP-2 [39] remains near 65.2%, showing only minor fluctuations. These results demonstrate that BayesMM is robust to hyperparameter variations and maintains consistent performance across models and datasets without fine-tuning.

Analysis of bayesian weighting. To better understand the role of modality balancing in BayesMM, we examine the learned Bayesian weights across different models on the ModelNet-C dataset. As shown in Figure 5, the textual modality consistently receives higher weights than the geometric one, reflecting its stronger stability and seman-

tic generalization under distribution shifts. This weighting pattern remains consistent across both clean and corrupted inputs, demonstrating that **BayesMM** adaptively calibrates cross-modal contributions and effectively suppresses noise sensitivity in the geometric stream, leading to more reliable multimodal inference.

4.4. Memory Usage and Throughput

Table 5. Memory usage (MB) comparison across different datasets. Numbers below each dataset name indicate the number of classes. Point-Cache denotes the hierarchical variant.

Method	ModelNet-C (40)	Omni3D (216)	O-LVIS (1156)	#Params (M)
ULIP [38]	1,556	1,558	1,556	85.7
+ Point-Cache	1,556	1,558	1,566	85.7
+ BayesMM	<u>1,560</u>	<u>1,560</u>	<u>1,562</u>	85.7
Uni3D [48]	5,062	5,062	5,062	1016.5
+ Point-Cache	<u>5,064</u>	<u>5,068</u>	5,090	1016.5
+ BayesMM	<u>5,076</u>	<u>5,077</u>	<u>5,080</u>	1016.5

Memory. We evaluate the memory consumption of our method compared with Point-Cache and the corresponding baselines ULIP [38] and Uni3D [48], as summarized in Table 5. Although our approach introduces a slightly higher memory footprint at ModelNet-C [21], its growth with respect to the number of categories is significantly slower. For example, when scaling from ModelNet-C (40 classes) [21] to O-LVIS (1,156 classes) [5], the total memory usage of Uni3D increases by nearly +18 MB under Point-Cache, whereas our hierarchical cache only adds about +4 MB. This trend demonstrates that our method effectively amortizes the class-wise parameter overhead by sharing covariance structures across categories. The result is a nearly constant per-class memory cost even in large-scale scenarios. Furthermore, the total memory remains dominated by the heavy backbone parameters of Uni3D ($\sim 1,016.5$ M), making the additional cost of our method negligible.

Throughput. We further compare inference throughput across different models. As shown in Table 6, **BayesMM** introduces only marginal overhead relative to Point-Cache and zero-shot baselines. Although additional refinement is performed, these operations are parallelizable on GPUs, yielding minimal runtime impact. Overall, our method preserves over 97% of the zero-shot inference speed while providing substantial gains in accuracy and robustness.

4.5. Visualization

To verify the effectiveness of **BayesMM**, we visualize feature distributions on ModelNet-C using t-SNE. As shown in Figure 6, the Cache-based updating strategy produces entangled clusters, showing that static cache features fail to capture discriminative semantics. Geometric distribu-

Table 6. Inference throughput (samples per second) on the ModelNet40. All experiments are conducted with a batch size of 1 on an RTX 3090 GPU. Point-Cache denotes the hierarchical variant.

Method	ULIP	ULIP2	OpenShape	Uni3D
Vanilla	11.35	12.82	8.68	7.77
+ Point-Cache [24]	<u>11.27</u>	<u>12.75</u>	<u>8.60</u>	<u>7.69</u>
+ BayesMM	10.99	12.45	8.29	7.45

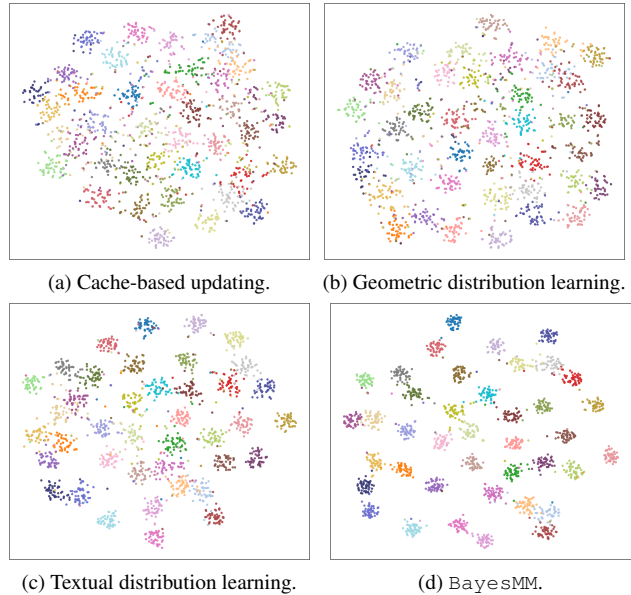


Figure 6. t-SNE visualization of feature distributions derived from classifiers constructed under different strategies. t-SNE visualization of feature distributions obtained from classifiers built with different learning strategies.

tion learning yields compact intra-class patterns and clear inter-class margins, indicating improved structural consistency. Textual distribution learning further aligns visual embeddings with textual priors, leading to better-separated clusters. Finally, **BayesMM** produces the most distinct and compact manifolds, as dynamically integrating geometric and textual evidences through Bayesian averaging reduces feature ambiguity and enhances discrimination.

5. Conclusion

This work presented **BayesMM**, a training-free dynamic Bayesian distribution learning framework for adaptive point cloud recognition under distribution shifts. By modeling geometric and textual modalities as Gaussian distributions and integrating them through Bayesian model averaging, **BayesMM** achieves stable and uncertainty-aware adaptation without additional training or auxiliary networks. Extensive experiments on corrupted, cross-domain, and large-scale benchmarks show that **BayesMM** substantially im-

proves robustness, stability, and cross-modal generalization compared with existing cache-based and test-time adaptation methods. Overall, *BayesMM* establishes a principled and efficient paradigm for multimodal test-time adaptation toward reliable 3D understanding in dynamic environments.

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A. Implementation Details

For ULIP [38] and ULIP-2 [39], we adopt PointBERT [41] as the point-cloud encoder backbone. For OpenShape [16], we follow the official configuration and use its scaled PointBERT variant with 32.1M parameters, as reported in Table J of the Appendix. For Uni3D, we employ the giant model, whose point encoder contains 1,016.5M parameters. All pretrained weights are obtained directly from their public GitHub repositories. We report the zero-shot recognition accuracy of these large 3D models as the baselines for comparison, and include Point-Cache [24] for completeness.

To describe point clouds, rather than relying on a single prompt such as “a point cloud object of a {class}”, we follow ULIP [38] and Point-PRC [23] and use 64 diverse text templates. Each template produces a textual description that is encoded into an embedding, and the 64 embeddings are averaged to obtain a class-level representation.

We further incorporate additional semantic detail only in the ModelNet-C evaluation. In this setting, the 64 generic templates are concatenated with 50 GPT-generated class-specific descriptions so that each category is provided with richer and more tailored semantics. For all subsequent evaluations, we use only the 64 generic templates without any class-specific augmentation. Even with this simplified setting, our method still yields competitive performance, which shows that it does not depend on complex or dataset-specific semantic information. This also indicates that the approach is easy to transfer across datasets without the need to generate separate prompts for each of them.

B. Derivation of Eq. (7) and Eq. (12)

B.1. Eq. (7): Textual prototype MAP

We derive the MAP estimator of the textual prototype ν_c in Eq. (7). For each class c , an LLM produces M paraphrased prompts with embeddings $\{\mathbf{z}_{c,1}, \dots, \mathbf{z}_{c,M}\}$, from which we compute the empirical mean and scatter:

$$\bar{\mathbf{z}}_c = \frac{1}{M} \sum_{i=1}^M \mathbf{z}_{c,i}, \quad \mathbf{S}_c = \sum_{i=1}^M (\mathbf{z}_{c,i} - \bar{\mathbf{z}}_c)(\mathbf{z}_{c,i} - \bar{\mathbf{z}}_c)^\top. \quad (15)$$

We treat $\bar{\mathbf{z}}_c$ as a sufficient statistic summarizing the M paraphrases and model the latent textual prototype ν_c as a Gaussian variable with prior

$$p(\nu_c) = \mathcal{N}(\nu_c \mid \mathbf{0}, \beta^2 \mathbf{I}), \quad (16)$$

where β^2 controls the prior variance.

Conditioned on ν_c , we assume that the empirical mean $\bar{\mathbf{z}}_c$ is drawn from a Gaussian whose covariance shrinks with the number of paraphrases:

$$p(\bar{\mathbf{z}}_c \mid \nu_c) = \mathcal{N}(\bar{\mathbf{z}}_c \mid \nu_c, \frac{1}{M} \mathbf{S}_c). \quad (17)$$

By Bayes’ rule, the posterior over ν_c is

$$p(\nu_c \mid \bar{\mathbf{z}}_c) \propto p(\bar{\mathbf{z}}_c \mid \nu_c) p(\nu_c). \quad (18)$$

Taking the negative log and omitting constants independent of ν_c gives

$$\begin{aligned} -\log p(\nu_c \mid \bar{\mathbf{z}}_c) &= \frac{1}{2\beta^2} \nu_c^\top \nu_c \\ &\quad + \frac{M}{2} (\bar{\mathbf{z}}_c - \nu_c)^\top \mathbf{S}_c^{-1} (\bar{\mathbf{z}}_c - \nu_c) \\ &\quad + \text{const}. \end{aligned} \quad (19)$$

Expanding the second term, we obtain:

$$\begin{aligned} -\log p(\nu_c \mid \bar{\mathbf{z}}_c) &= \frac{1}{2} \nu_c^\top (\beta^{-2} \mathbf{I} + M \mathbf{S}_c^{-1}) \nu_c \\ &\quad - \nu_c^\top (M \mathbf{S}_c^{-1} \bar{\mathbf{z}}_c) + \text{const}, \end{aligned} \quad (20)$$

which matches the canonical Gaussian form in ν_c with precision

$$\Lambda_c = \beta^{-2} \mathbf{I} + M \mathbf{S}_c^{-1}, \quad (21)$$

and natural parameter

$$\eta_c = M \mathbf{S}_c^{-1} \bar{\mathbf{z}}_c. \quad (22)$$

Thus the posterior over ν_c is Gaussian,

$$p(\nu_c \mid \bar{\mathbf{z}}_c) = \mathcal{N}(\nu_c \mid \nu_c^{\text{MAP}}, \Sigma_{\nu_c}), \quad (23)$$

with

$$\Sigma_{\nu_c} = \Lambda_c^{-1} = (\beta^{-2} \mathbf{I} + M \mathbf{S}_c^{-1})^{-1}, \quad (24)$$

$$\nu_c^{\text{MAP}} = \Sigma_{\nu_c} \eta_c = (\beta^{-2} \mathbf{I} + M \mathbf{S}_c^{-1})^{-1} M \mathbf{S}_c^{-1} \bar{\mathbf{z}}_c. \quad (25)$$

Since $\mathbf{S}_c$ in Eq. (3) is an unnormalized scatter matrix, its global scale can be absorbed into M without changing the relative weighting between the prior and data terms. Under this convention, simplifying the common scalar factor yields the compact expression used in the main paper:

$$\nu_c^{\text{MAP}} = (\beta^{-2} \mathbf{I} + M \mathbf{S}_c^{-1})^{-1} \mathbf{S}_c^{-1} \bar{\mathbf{z}}_c, \quad (26)$$

which gives Eq. (7).

B.2. Eq. (12): Geometric distribution update

We next derive the recursive update in Eq. (12) for a fixed class c , omitting the class index when unambiguous. At test-time step t , the geometric parameters are:

$$\Theta_t = \{\mu_t, \Sigma_t\}, \quad (27)$$

and the prior is given by the previous posterior:

$$p(\Theta_t) = p(\Theta_{t-1} \mid \mathbf{x}_{t-1}). \quad (28)$$

Under the Gaussian assumptions in Eq. (8) and Eq. (10), the mean evolves according to:

$$\boldsymbol{\mu}_t \sim \mathcal{N}(\boldsymbol{\mu}_{t-1}, \boldsymbol{\Sigma}_{t-1}), \quad (29)$$

$$\mathbf{x}_t \mid \boldsymbol{\mu}_t \sim \mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}). \quad (30)$$

Applying Bayes' rule yields:

$$p(\boldsymbol{\mu}_t \mid \mathbf{x}_t) \propto p(\mathbf{x}_t \mid \boldsymbol{\mu}_t) p(\boldsymbol{\mu}_t). \quad (31)$$

Taking the negative log (up to constants independent of $\boldsymbol{\mu}_t$) gives:

$$\begin{aligned} -\log p(\boldsymbol{\mu}_t \mid \mathbf{x}_t) &= \frac{1}{2}(\boldsymbol{\mu}_t - \boldsymbol{\mu}_{t-1})^\top \boldsymbol{\Sigma}_{t-1}^{-1}(\boldsymbol{\mu}_t - \boldsymbol{\mu}_{t-1}) \\ &\quad + \frac{1}{2}(\mathbf{x}_t - \boldsymbol{\mu}_t)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}_t - \boldsymbol{\mu}_t). \end{aligned} \quad (32)$$

Expanding the quadratic terms leads to:

$$\begin{aligned} -\log p(\boldsymbol{\mu}_t \mid \mathbf{x}_t) &= \frac{1}{2}\boldsymbol{\mu}_t^\top (\boldsymbol{\Sigma}_{t-1}^{-1} + \boldsymbol{\Sigma}^{-1})\boldsymbol{\mu}_t \\ &\quad - \boldsymbol{\mu}_t^\top (\boldsymbol{\Sigma}_{t-1}^{-1}\boldsymbol{\mu}_{t-1} + \boldsymbol{\Sigma}^{-1}\mathbf{x}_t) + \text{const}, \end{aligned} \quad (33)$$

which matches the canonical Gaussian form with precision:

$$\boldsymbol{\Lambda}_t = \boldsymbol{\Sigma}_{t-1}^{-1} + \boldsymbol{\Sigma}^{-1}, \quad (34)$$

and natural parameter:

$$\boldsymbol{\eta}_t = \boldsymbol{\Sigma}_{t-1}^{-1}\boldsymbol{\mu}_{t-1} + \boldsymbol{\Sigma}^{-1}\mathbf{x}_t. \quad (35)$$

Thus the posterior is Gaussian with parameters:

$$\boldsymbol{\Sigma}_t = \boldsymbol{\Lambda}_t^{-1} = (\boldsymbol{\Sigma}_{t-1}^{-1} + \boldsymbol{\Sigma}^{-1})^{-1}, \quad (36)$$

$$\boldsymbol{\mu}_t = \boldsymbol{\Sigma}_t \boldsymbol{\eta}_t = \boldsymbol{\Sigma}_t (\boldsymbol{\Sigma}_{t-1}^{-1}\boldsymbol{\mu}_{t-1} + \boldsymbol{\Sigma}^{-1}\mathbf{x}_t). \quad (37)$$

Restoring the class index and substituting the class-specific covariance $\boldsymbol{\Sigma}^c$, we obtain:

$$\boldsymbol{\mu}_t^c = \boldsymbol{\Sigma}_t^c \left[(\boldsymbol{\Sigma}^c)^{-1}\mathbf{x}_t + (\boldsymbol{\Sigma}_{t-1}^c)^{-1}\boldsymbol{\mu}_{t-1}^c \right], \quad (38)$$

$$\boldsymbol{\Sigma}_t^c = \left[(\boldsymbol{\Sigma}_{t-1}^c)^{-1} + (\boldsymbol{\Sigma}^c)^{-1} \right]^{-1}, \quad (39)$$

which gives Eq. (12) in the main paper.

C. Additional Results

Robustness evaluation. Tables G, H, and I present the recognition accuracy under different corruption settings. Overall, BayesMM consistently improves performance over the baseline models (ULIP, ULIP-2, O-Shape, and Uni3D) as well as the Point-Cache variants.

On S-OBJ_ONLY-C (Table G), BayesMM consistently outperforms Point-Cache, improving the average accuracy by 3.0% to 4.5%, and achieves a 4% to 7% gain over the

original backbone models, demonstrating clear improvements in robustness. On S-OBJ_BG-C (Table H), which introduces background clutter, BayesMM achieves 2.5% to 3.5% higher average accuracy than Point-Cache and up to 6% to 8% improvement over the original models, indicating strong generalization in more challenging scenes. On the most challenging split S-PB.T50-RS-C (Table I), BayesMM increases the average accuracy by 4.5% to 5.0% over Point-Cache and by 7% to 10% over the original backbones, demonstrating that our approach can effectively handle severe corruptions and partial observations, significantly enhancing model robustness. These consistent gains across settings highlight the effectiveness of BayesMM in improving robustness under diverse and severe corruptions.

Memory usage and inference throughput. Table J reports the memory consumption across ModelNet-C, Omni3D, and O-LVIS. While Point-Cache exhibits comparable or slightly higher memory usage on smaller datasets, its parameter footprint grows substantially as the number of object categories increases, particularly on O-LVIS. In contrast, BayesMM maintains a consistently lightweight profile, indicating that the robustness improvements introduced by our method come with only negligible memory overhead. Table K presents the inference throughput measured on S-OBJ_ONLY. The throughput reduction introduced by BayesMM remains within 2% to 5% across all evaluated backbones, indicating that the additional operations do not significantly affect runtime. The resulting throughput decrease remains modest relative to the robustness enhancements achieved by BayesMM.

Table G. Comparison of recognition accuracy on S-OBJ-ONLY-C, which contains seven types of corruptions. Results are reported at corruption severity level 2. Each clean point cloud contains 1024 points. SONN refers to ScanObjectNN.

Method	Original Data	Corruption Type							Avg.
	SONN	Add Global	Add Local	Drop Global	Drop Local	Rotate	Scale	Jitter	
ULIP [38]	49.05	31.50	34.77	51.29	38.38	48.36	44.58	36.83	41.85
+ Point-Cache (Global)	<u>52.15</u>	<u>35.80</u>	<u>37.01</u>	<u>54.39</u>	<u>41.82</u>	<u>49.74</u>	<u>45.09</u>	40.28	44.54
+ Point-Cache (Hierarchical)	52.15	32.01	38.04	<u>54.56</u>	<u>45.27</u>	<u>50.95</u>	<u>45.96</u>	<u>39.24</u>	<u>44.77</u>
+ BayesMM	54.04	39.24	38.04	55.42	46.30	52.15	47.68	35.11	45.75
ULIP-2 [39]	42.00	40.45	41.31	37.69	30.29	38.21	44.45	22.89	37.16
+ Point-Cache (Global)	48.19	<u>49.05</u>	<u>46.30</u>	<u>45.09</u>	<u>37.18</u>	<u>41.65</u>	<u>44.41</u>	<u>25.99</u>	42.24
+ Point-Cache (Hierarchical)	<u>51.98</u>	<u>49.05</u>	<u>46.30</u>	48.88	<u>40.45</u>	45.78	<u>45.09</u>	<u>25.99</u>	<u>44.19</u>
+ BayesMM	52.67	54.04	48.02	<u>45.78</u>	42.34	<u>44.75</u>	45.96	28.57	45.52
O-Shape [16]	53.18	49.91	46.30	52.15	36.66	46.64	46.82	30.81	45.31
+ Point-Cache (Global)	56.80	56.45	51.98	54.56	40.45	<u>51.81</u>	<u>49.23</u>	37.69	49.90
+ Point-Cache (Hierarchical)	<u>58.69</u>	<u>59.04</u>	<u>53.01</u>	<u>55.94</u>	<u>41.82</u>	<u>51.12</u>	<u>48.54</u>	<u>39.41</u>	<u>50.95</u>
+ BayesMM	61.96	61.10	55.76	59.03	48.54	54.90	53.35	40.96	54.44
Uni3D [48]	65.58	62.65	56.45	60.07	49.40	61.62	56.11	43.55	56.93
+ Point-Cache (Global)	70.05	65.06	<u>59.38</u>	63.68	54.39	<u>63.34</u>	60.07	51.29	60.91
+ Point-Cache (Hierarchical)	70.22	65.40	58.00	<u>64.20</u>	<u>54.91</u>	61.96	62.13	53.18	61.25
+ BayesMM	71.60	69.53	60.06	64.54	60.07	68.33	63.86	52.32	63.79

Table H. Comparison of recognition accuracy on S-OBJ-BG-C, which includes seven types of corruptions. Results are reported at corruption severity level 2. Each clean point cloud contains 1024 points.

Method	Original Data	Corruption Type							Avg.
	SONN	Add Global	Add Local	Drop Global	Drop Local	Rotate	Scale	Jitter	
ULIP [38]	45.96	27.19	25.82	45.61	34.25	40.96	40.10	30.98	36.36
+ Point-Cache (Global)	48.88	30.46	<u>30.46</u>	<u>49.05</u>	39.59	<u>44.92</u>	<u>42.17</u>	<u>31.84</u>	<u>39.68</u>
+ Point-Cache (Hierarchical)	<u>49.74</u>	28.23	30.12	48.71	<u>40.45</u>	43.55	40.28	34.42	39.44
+ BayesMM	52.67	33.05	34.08	50.43	41.65	48.53	45.78	31.50	41.81
ULIP-2 [39]	48.19	40.62	38.90	39.24	32.36	41.14	42.86	21.17	38.04
+ Point-Cache (Global)	52.50	48.19	45.09	46.82	39.07	46.64	48.02	26.51	44.10
+ Point-Cache (Hierarchical)	<u>54.73</u>	51.64	47.16	50.95	<u>39.76</u>	53.01	51.81	22.72	46.47
+ BayesMM	56.80	<u>50.77</u>	<u>46.82</u>	<u>49.40</u>	40.45	<u>50.26</u>	<u>49.57</u>	<u>25.47</u>	<u>46.19</u>
O-Shape [16]	55.94	49.40	48.19	52.67	42.51	48.88	47.16	31.84	47.08
+ Point-Cache (Global)	59.72	57.49	51.12	<u>59.72</u>	<u>48.71</u>	<u>56.11</u>	<u>54.22</u>	35.28	52.80
+ Point-Cache (Hierarchical)	<u>62.65</u>	<u>58.00</u>	<u>51.64</u>	<u>59.55</u>	<u>47.85</u>	<u>54.91</u>	<u>53.36</u>	<u>36.49</u>	<u>53.06</u>
+ BayesMM	64.72	60.41	54.90	61.62	52.32	60.41	57.14	38.21	55.09
Uni3D [48]	60.24	58.00	52.32	51.64	44.23	58.00	51.81	39.24	51.94
+ Point-Cache (Global)	<u>63.86</u>	66.27	57.83	56.11	<u>50.77</u>	<u>61.62</u>	56.11	44.23	57.10
+ Point-Cache (Hierarchical)	62.82	64.72	<u>57.14</u>	<u>58.52</u>	50.43	60.93	59.55	<u>46.30</u>	<u>57.55</u>
+ BayesMM	68.50	<u>66.26</u>	54.39	60.58	55.07	65.23	<u>58.86</u>	49.57	59.06

Table I. Comparison of corruption generalization on S-PB_T50-RS-C, the most challenging split of ScanObjectNN. Each clean point cloud is represented by 1024 points. SONN denotes ScanObjectNN.

Method	Original Data	Corruption Type							Avg.
	SONN	Add Global	Add Local	Drop Global	Drop Local	Rotate	Scale	Jitter	
ULIP [38]	29.29	19.26	18.39	30.99	23.91	27.48	26.34	21.44	24.64
+ Point-Cache (Global)	32.37	22.87	20.85	33.31	27.90	30.85	<u>28.63</u>	24.53	27.66
+ Point-Cache (Hierarchical)	<u>32.48</u>	<u>23.46</u>	<u>22.69</u>	<u>34.70</u>	<u>31.75</u>	<u>33.00</u>	28.28	<u>25.05</u>	<u>28.93</u>
+ BayesMM	40.52	29.53	25.92	39.21	33.59	35.74	32.44	24.67	33.18
ULIP-2 [39]	33.38	30.29	29.42	28.24	24.91	28.56	30.22	12.98	27.25
+ Point-Cache (Global)	40.28	<u>36.40</u>	33.80	35.39	30.88	33.66	35.01	18.36	32.97
+ Point-Cache (Hierarchical)	<u>42.40</u>	35.70	<u>34.42</u>	<u>37.75</u>	<u>34.21</u>	<u>36.26</u>	<u>36.09</u>	<u>19.12</u>	<u>34.49</u>
+ BayesMM	46.31	41.29	37.82	40.46	34.57	39.73	37.51	16.53	36.78
O-Shape [16]	41.12	32.41	35.60	37.80	27.34	36.61	35.22	18.88	33.12
+ Point-Cache (Global)	42.16	40.32	37.58	42.02	33.76	41.53	<u>38.24</u>	24.12	37.47
+ Point-Cache (Hierarchical)	<u>43.72</u>	<u>40.91</u>	<u>39.24</u>	<u>43.03</u>	<u>35.22</u>	<u>43.06</u>	37.40	<u>25.05</u>	<u>38.45</u>
+ BayesMM	50.52	49.51	43.64	49.53	41.22	47.11	45.31	30.29	44.39
Uni3D [48]	46.04	48.23	37.99	36.75	31.47	44.00	37.37	28.66	38.46
+ Point-Cache (Global)	50.28	<u>52.57</u>	<u>42.23</u>	42.61	36.29	47.22	39.83	33.48	43.06
+ Point-Cache (Hierarchical)	<u>51.13</u>	51.67	41.88	<u>44.59</u>	<u>38.79</u>	<u>49.03</u>	<u>41.05</u>	<u>34.70</u>	<u>44.10</u>
+ BayesMM	57.04	59.30	45.70	49.10	44.41	53.16	48.37	37.86	49.17

Table J. Memory usage (MB) comparison across different datasets. Numbers below each dataset name indicate the number of classes. Point-Cache denotes the hierarchical variant.

Method	ModelNet-C (40)	Omni3D (216)	O-LVIS (1156)	#Params (M)
ULIP-2	1,556	1,558	1,556	85.7
+ Point-Cache	1,556	1,558	1,566	85.7
+ BayesMM	<u>1,560</u>	<u>1,560</u>	<u>1,562</u>	85.7
OpenShape	7,056	7,058	7,116	2,571.9
+ Point-Cache	<u>7,058</u>	<u>7,062</u>	<u>7,150</u>	2,571.9
+ BayesMM	<u>7,076</u>	<u>7,080</u>	<u>7,084</u>	2,571.9

Table K. Inference throughput (samples per second) on S-OBJ_ONLY. All experiments are conducted with a batch size of 1 on an RTX 3090 GPU. Point-Cache refers to the hierarchical variant of Point-Cache here.

Method	ULIP	ULIP2	OpenShape	Uni3D
Vanilla	11.25	11.25	8.60	7.72
+ Point-Cache [24]	<u>11.17</u>	<u>11.17</u>	<u>8.57</u>	<u>7.62</u>
+ BayesMM	10.90	10.91	8.28	7.41